FED. SUP CLASS
53GP

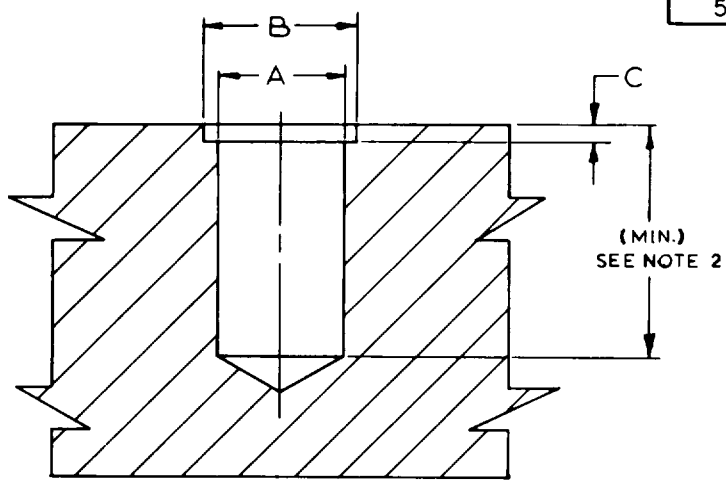


FIGURE I

1. AXIS OF HOLE TO BE NORMAL TO ENTRY SURFACE OR PROVIDE SPOTFACE WHEN REQUIRED.
2. MINIMUM HOLE DEPTH TO BE LENGTH OF INSERT PLUS .06.
3. SURFACE ROUGHNESS: UNLESS OTHERWISE SPECIFIED, MACHINED SURFACES TO BE 125 MICROINCHES IN ACCORDANCE WITH USA STANDARD B46.1.
4. REMOVE ALL BURRS AND SHARP EDGES.
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES, LINEAR DIMENSIONS ± .005.

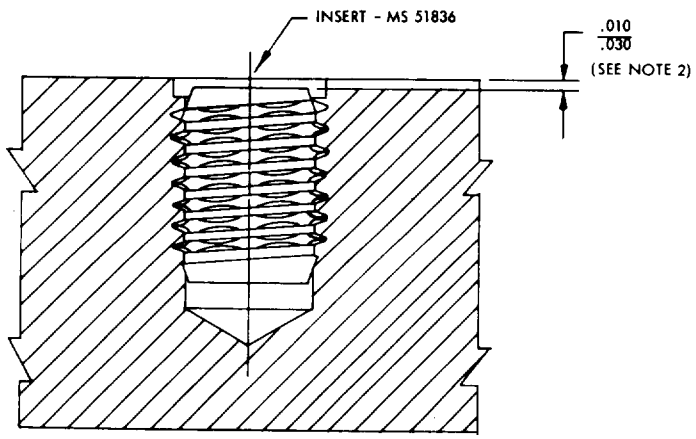


FIGURE II

1. HOLE PREPARATION IN ACCORDANCE WITH FIGURE I AND TABLE I.
2. INSTALL INSERT TO DEPTH SHOWN.
3. TYPICAL DRAWING CALLOUT TO BE LOCATED IN VICINITY OF PART IDENTIFICATION.



INSERT VIEW

THIS IS A DESIGN STANDARD AND SHALL NOT BE USED AS A PART NUMBER.

REVIEWER: ML, AT, ME, AV, IS, NSA
USER: EL, GL, YD, MC

This military standard has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 1 DEC 1967 REVISED

P.A. WC Other Cost AS 82	TITLE INSERT, SCREW THREAD - THREAD FORMING HOLE DIMENSIONS FOR AND INSTALLATION OF	MILITARY STANDARD MS 51837
PROCUREMENT SPECIFICATION NOT APPLICABLE	SUPERSEDES:	SHEET 1 OF 2

DD FORM 1 SEP 67 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:10Z
Check the source to verify that this is the current version before use.

TABLE I

DRILLED HOLE SIZES FOR MS 51836 INSERTS

NOMINAL INTERNAL THREAD SIZE (MS 51836) (REF)	GROUP A MATERIALS			GROUP B MATERIALS			GROUP C MATERIALS		
	A DIA	B DIA	C ±.010	A DIA	B DIA	C ±.010	A DIA	B DIA	C ±.010
.086-56 .086-64	.121 .118	.162 .158	.020	.127 .124	.162 .158	.020	.127 .124	.162 .158	.020
.112-40 .112-48	.150 .146	.173 .169	.040	.162 .158	.180 .176	.040	.164 .160	.180 .176	.040
.138-32 .138-40	.191 .187	.216 .212	.040	.204 .200	.224 .220	.040	.209 .205	.224 .220	.040
.164-32 .164-36	.224 .220	.254 .249	.040	.240 .235	.265 .260	.040	.244 .239	.265 .260	.040
.190-24 .190-32	.243 .238	.285 .280	.050	.262 .257	.294 .289	.050	.264 .259	.294 .289	.050
.250-20 .250-28	.332 .327	.372 .367	.050	.348 .343	.379 .374	.050	.352 .347	.379 .374	.050
.3125-18 .3125-24	.403 .398	.472 .467	.050	.417 .412	.472 .467	.050	.425 .420	.472 .467	.050
.375-16 .375-24	.488 .483	.567 .562	.060	.521 .515	.568 .562	.060	.531 .526	.568 .562	.060

MATERIAL GROUPS: GROUP A - SOFT PLASTICS

GROUP B - CAST ALUMINUM, MAGNESIUM AND HARD PLASTICS

GROUP C - HIGH STRENGTH ALUMINUM ALLOYS (EXAMPLE: 2024-T4)

This military standard has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEWER: MI, AT, ME, AV, IS, NSA, EL, GL, YD, MC
USER:

APPROVED 1 DEC 1987 REVISED

P.A. WC Other Cust AS 82	TITLE INSERT, SCREW THREAD - THREAD FORMING HOLE DIMENSIONS FOR AND INSTALLATION OF	MILITARY STANDARD MS 51837
PROCUREMENT SPECIFICATION NOT APPLICABLE	SUPERSEDES	SHEET 2 OF

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:10Z
Check the source to verify that this is the current version before use.